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ADIS DRUG EVALUATION

Aumolertinib: A Review in Non-Small Cell Lung Cancer

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Abstract

Aumolertinib (formerly almonertinib; Ameile®) is an oral, third-generation epidermal growth factor receptor (EGFR) tyrosine kinase inhibitor (EGFR-TKI) that is selective for mutant EGFR over wild-type EGFR. It has been developed for the treatment of advanced EGFR mutation-positive non-small cell lung cancer (NSCLC). In the phase 3 AENEAS trial conducted in Chinese patients, aumolertinib as first-line treatment significantly prolonged progression-free survival (PFS) and duration of response (DoR) compared with gefitinib in patients with advanced EGFR mutation-positive NSCLC; overall survival (OS) data from this study are immature. In the phase 1/2 APOLLO trial, aumolertinib showed good clinical activity (based on objective response rate, PFS, DoR and OS) in Chinese patients with locally advanced or metastatic EGFR T790M mutation-positive NSCLC who had progressed on or after prior EGFR-TKI therapy. Aumolertinib has a generally manageable tolerability profile; adverse events associated with wild-type EGFR inhibition (e.g. rash and diarrhoea) were less frequent with aumolertinib than gefitinib in AENEAS. Thus, aumolertinib is a promising new option for both first-line and second-line treatment in patients with advanced EGFR mutation-positive NSCLC.

Plain Language Summary

Non-small cell lung cancer (NSCLC), the most common type of lung cancer, is associated with a poor prognosis. In up to 50% of Asian patients and ≈ 17% of Caucasian patients with NSCLC, specific mutations in the epidermal growth factor receptor (EGFR) gene are responsible for tumour growth. Drugs that block the effects of EGFR [known as EGFR tyrosine kinase inhibitors (EGFR-TKIs)] improve survival in patients with NSCLC and EGFR mutations. Aumolertinib (Ameile®) is a third-generation EGFR-TKI that has been developed in China. When given to patients with previously untreated, advanced EGFR mutation-positive NSCLC, aumolertinib delayed cancer progression by ≈ 9 months compared with gefitinib, a first-generation EGFR-TKI. Aumolertinib also showed good clinical activity as a second-line treatment in patients with advanced EGFR mutation-positive NSCLC who had developed resistance during or after treatment with earlier-generation EGFR-TKIs. Overall, the tolerability profile of aumolertinib was similar to that of gefitinib, with some adverse events such as rash and diarrhoea being less frequent. Aumolertinib is a promising new first- or second-line treatment option in patients with advanced EGFR mutation-positive NSCLC.

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Aumolertinib: clinical considerations in advanced EGFR mutation-positive NSCLC

- Orally administered third-generation EGFR-TKI that has been developed in China
- Selective for mutant EGFR over wild-type EGFR
- Has high inhibitory activity against EGFR with common sensitizing mutations and the T790M resistance mutation
- Prolongs PFS and DoR relative to gefitinib as first-line treatment; good clinical activity (based on objective response rate, PFS, DoR and OS) as second-line treatment
- Generally manageable tolerability profile similar to other EGFR-TKIs

1 Introduction

The first-generation epidermal growth factor receptor (EGFR) tyrosine kinase inhibitors (TKIs) have been shown to provide significant clinical benefit in the treatment of patients with advanced non-small cell lung cancer (NSCLC) with EGFR sensitizing mutations [exon 19 deletions (ex19del) and exon 21 L858R mutations] and are established as valid options for first-line therapy [1–4]. However, after a median duration of response (DoR) of ≈ 12 months, disease progression occurs following the emergence of tumour resistance mutations (most commonly a T790M mutation) [5–7]. A recent analysis of nearly 17,000 patients with EGFR-mutant NSCLC found that 67.1% had classical EGFR mutations (L858R and/or ex19del with or without T790M) and 30.8% had atypical EGFR mutations [8]. Although effective in their activity against T790M mutation-positive EGFR, second-generation EGFR-TKIs developed to treat resistant disease have largely been limited by dose-limiting toxicity due to their inhibitory activity against wild-type EGFR [9–12]. Third-generation EGFR-TKIs (of which osimertinib was the first to be widely approved [13–16]) irreversibly bind and inhibit EGFR with sensitizing mutations and T790M resistance mutations [17, 18]. Furthermore, the enhanced selectivity of third-generation EGFR-TKIs for mutated EGFR over wild-type EGFR is designed to improve their tolerability [19–21].

Aumolertinib (formerly almonertinib; Ameile®) (Fig. 1) is one of the most recent third-generation EGFR-TKIs to be approved for the treatment of locally advanced or metastatic EGFR mutation-positive NSCLC [14, 22, 23]; among the other EGFR-TKIs in clinical development in this indication, lazertinib and furmonertinib are now approved in South Korea and China, respectively [14, 24, 25]. Aumolertinib is indicated in China as first-line treatment for adults with untreated advanced NSCLC and EGFR ex19del or L858R mutations [26] as well as for use in adult patients with EGFR T790M mutation-positive NSCLC [confirmed using a National Medical Products Administration (NMPA)-approved test] who have progressed on or after prior EGFR-TKI therapy [23]. This article reviews the clinical efficacy and tolerability of aumolertinib in the management of NSCLC with EGFR sensitizing and resistance mutations and summarizes the pharmacological properties of the drug.

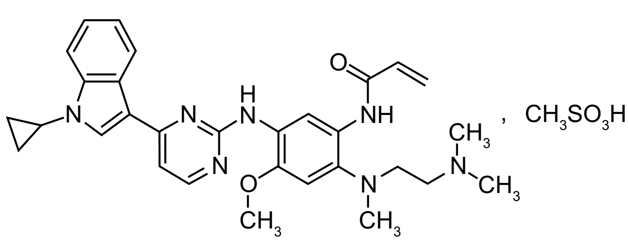

Fig. 1 Chemical structure of aumolertinib

2 Pharmacological Properties of Aumolertinib

2.1 Pharmacodynamics

Aumolertinib is an irreversible, small-molecule EGFR-TKI with high inhibitory activity against EGFR with common sensitizing mutations (ex19del and L858R) as well as the T790M resistance mutation [23, 27, 28]. Aumolertinib is highly selective, having IC_{50} values for EGFR with resistance or sensitizing mutations that are approximately 2- to 16-fold lower than that for the wild-type enzyme [23, 28]. The in vitro activity and selectivity of HAS-719, the active N-desmethyl metabolite of aumolertinib (Sect. 2.2), was similar to that of aumolertinib against EGFR resistance or sensitizing mutations. The activity and selectivity of aumolertinib in enzymatic assays is comparable to that of osimertinib [28]. In cell proliferation assays, aumolertinib displayed potent activity comparable to that of osimertinib in cell lines harbouring EGFR sensitizing and resistance mutations (ex19del, L858R, T790M) (Fig. 2) and comparable to that of gefitinib in cell lines with ex19del mutation (gefitinib showed no activity against T790M mutated cells) [23, 28]. In addition, the anti-tumour activity of aumolertinib has been demonstrated using patient-derived xenograft (PDX) models of NSCLC [23, 28], as well as in the phase 1/2

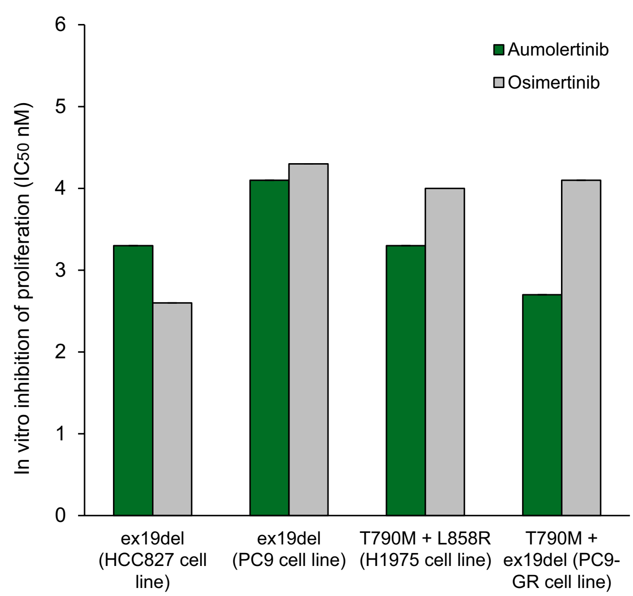

Fig. 2 In vitro activity of aumolertinib and osimertinib based on cell proliferation assays [28]. *ex19del* exon 19 deletion

APOLLO clinical trial [27, 29, 30]. Aumolertinib showed dose-dependent anti-tumour activity in PDX tumour models with EGFR T790M and sensitizing mutations, comparable to that of osimertinib [28].

Mechanisms of resistance to aumolertinib identified in patients with disease progression in the APOLLO trial included EGFR C797S and EGFR L718Q mutations and aberrations in bypass tracts (including PIK3CA, JAK2, BRAF and KRAS mutations, HER2 amplification and FGFR3-TACC3 fusion) [30].

2.2 Pharmacokinetics

Aumolertinib exposure increased linearly in a dose-dependent manner over the dose range of 55–220 mg, with evidence of saturation at a dose of 260 mg [27]. After a single oral dose of aumolertinib 110 mg in patients with advanced NSCLC, aumolertinib reached a mean peak plasma concentration (C_{max}) of 318.5 ng/mL in a median time of 4 h [23]. Mean C_{max} of the active metabolite HAS-719 was reached in a median time of 17–18 h. Administration of aumolertinib with a high-fat meal had no clinically significant effects on aumolertinib and HAS-719 pharmacokinetics. At steady state, aumolertinib and HAS-719 had mean accumulation ratios of 1.39 and 4.07, respectively. Aumolertinib is widely distributed in the body (apparent volume of distribution = 554 L). In vitro, aumolertinib and HAS-719 are highly (≥ 99.5%) bound by human plasma proteins [23].

Aumolertinib metabolism primarily occurs via CYP3A4. Co-administration of aumolertinib with strong CYP3A4 inhibitors or inducers can result in significant increases or reductions in aumolertinib exposure. In patients with advanced NSCLC, the mean terminal elimination half-lives of aumolertinib and HAS-719, respectively, are 30.6 h and 55.4 h. Drug excretion primarily occurs via the faecal route (85%); 5% is excreted in urine [23].

Based on drug concentrations in brain tissue and plasma, aumolertinib crossed the blood-brain barrier after oral administration in a brain metastasis xenograft mouse model; however, the concentration of the active metabolite HAS-719 in brain tissue was negligible compared with that detected in plasma [31]. Aumolertinib also inhibited intracranial and spinal cord tumour growth in brain metastasis and spinal cord metastasis mouse models [31] and achieved similar exposure to osimertinib in mouse brain tissues after a single 5 mg/kg dose of either drug [28].

No aumolertinib dose adjustments are necessary based on patient age, bodyweight or gender or for mild hepatic impairment or mild to moderate renal impairment. The effects of moderate to severe hepatic impairment, severe renal impairment or end-stage renal disease on aumolertinib pharmacokinetics have not been determined [23].

3 Therapeutic Efficacy of Aumolertinib

The efficacy of oral aumolertinib in patients with EGFR mutation-positive, advanced NSCLC has been investigated as first-line treatment in the randomized, open-label, active-controlled, phase 3 AENEAS trial (Sect. 3.1) [32] and as second-line treatment in the single-arm phase 1/2 APOLLO trial (Sect. 3.2) [27, 29, 30]. The primary efficacy endpoint in AENEAS was investigator-assessed progression-free survival (PFS) by RECIST v1.1 criteria [32]. The trial had 90% power to detect a PFS hazard ratio (HR) of 0.67 at 262 PFS events [32]. The primary efficacy endpoint in the phase 2 part of APOLLO was the objective response rate (ORR) [27, 29, 30]. Both trials were conducted in China.

3.1 First-Line Monotherapy in EGFR Mutation-Positive NSCLC

Preliminary results of the AENEAS trial indicate that as first-line treatment in patients with EGFR mutation-positive, advanced (stage IIIB/IV) NSCLC, aumolertinib is more effective than gefitinib in prolonging PFS. In this trial, 429 patients with untreated advanced NSCLC and EGFR ex19del or L858R mutations were randomized to receive either aumolertinib 110 mg once daily (n = 214) or gefitinib 250 mg once daily (n = 215) [32]. The demographic characteristics of patients at baseline were similar in the two treatment arms. Patients were stratified by mutation status (ex19del or L858R) and whether or not brain metastasis was present at baseline. Gefitinib recipients were eligible to cross over to aumolertinib if they developed a T790M resistance mutation on local or central testing. [32].

At data cut-off (planned final event-driven PFS analysis at 262 PFS events), aumolertinib recipients had a significantly (p < 0.0001) longer PFS than gefitinib recipients [based on assessment by independent central review (ICR) every 6 weeks] and a significantly longer DoR (Table 1) [32]. Aumolertinib was also significantly more effective than gefitinib in prolonging PFS in the subgroup of patients with CNS metastases at baseline (HR 0.38; 95% CI 0.24–0.60). Other subgroup analyses showed that regardless of EGFR mutation type, sex, age, smoking history and ECOG performance status score, PFS was significantly prolonged with aumolertinib compared with gefitinib. OS data in both treatment arms were immature at data cut-off [32].

3.2 Second-Line Monotherapy in EGFR Mutation-Positive NSCLC

Aumolertinib has good clinical activity in patients with locally advanced or metastatic (stage IIIB/IV), EGFR

Table 1 Efficacy of aumolertinib in patients with locally advanced or metastatic NSCLC with EGFR sensitizing or resistance mutations

Treatment[a] (no. of patients)	Median PFS (mo)	Median OS (mo)	Median DoR (mo)	ORR (%)
First-line treatment in patients with EGFR ex19del or L858R mutation-positive NSCLC (phase 3 AENEAS trial) [32]				
Aumolertinib 110 mg (214)	19.3[b]	Not mature	18.1	73.8
Gefitinib 250 mg (215)	9.9[b]	Not mature	8.3	72.1
	[HR 0.46 (95% CI 0.36–0.60)]*		[HR 0.38 (95% CI 0.28–0.51]*	[OR 1.09 (95% CI 0.70–1.69]
Second-line treatment in patients with EGFR T790M mutation-positive NSCLC that has progressed after first- or second-generation EGFR-TKI treatment (phase 2 data from the phase 1/2 APOLLO trial) [29, 30]				
Aumolertinib 110 mg (244)	12.4	30.2[c]	15.1	68.9[b]

DoR duration of response, *EGFR* epidermal growth factor receptor, *ex19del* exon 19 deletion, *HR* hazard ratio, *mo* months *NSCLC* non-small cell lung cancer, *OR* odds ratio, *ORR* objective response rate, *OS* overall survival, *PFS* progression-free survival, *TKI* tyrosine kinase inhibitor

*$p < 0.0001$ vs gefitinib

[a]Administered orally once daily until disease progression, unacceptable toxicity or patient withdrawal

[b]Primary endpoint. Evaluated at 262 PFS events in AENEAS and at a median follow-up of 19.4 mo in APOLLO

[c]Secondary endpoint. Evaluated at a median follow-up of 34.5 mo

T790M mutation-positive NSCLC that has progressed after first- or second-generation EGFR-TKI treatment, according to data from the phase 1/2 APOLLO trial [27, 29, 30]. In the phase 2 part of the trial, 244 adult patients were treated with aumolertinib 110 mg once daily until disease progression, unacceptable toxicity or patient withdrawal [29, 30]. Patients with asymptomatic, stable brain metastases not requiring steroids were eligible to enrol; these patients comprised 36.1% of the full analysis set [29, 30].

At data cut-off (median follow-up of 19.4 months), the ORR was 68.9%, based on assessment by ICR every 6 weeks using RECIST v1.1 criteria [29]. The disease control rate (DCR; defined as the percentage of patients with a complete or partial response or stable disease) was 93.4%. Patients had a median PFS of just over 12 months and the median DoR among responders was just over 15 months (Table 1). When evaluated according to baseline mutation (ex19del or L858R), efficacy outcomes were comparable with those in the overall population [29]. In the subset of 88 patients with brain metastases at baseline, 23 had measurable lesions that were evaluated by ICR. Among these patients, the ORR was 60.9% (95% CI 38.5–80.3), DCR was 91.3%, the median PFS was 11.8 months and the median DOR was 12.5 months (median follow-up 17.9 months). One of these patients had a CNS complete response [29].

At a subsequent data cut-off (20 July 2021; final OS evaluation at a median follow-up of 34.5 months), the median OS was 30.2 months, and the OS rate at 24 months was 57.5% [30]. At this assessment, just over half of the patients in the trial had died (126/244; 51.6%). OS in patients with brain metastases was 19.1 months, and that in patients without brain metastases was 36.4 months [30].

Patients in the phase 2 part of the trial had a median age of 61 years (range, 27–87 years) and an ECOG score of 0 (34.8% of patients) or 1 (65.2%); all were of Asian race [29, 30]. The NSCLC was classified as adenocarcinoma in 99.2% of patients and squamous cell carcinoma in 0.8% of patients. T790M mutations were associated with ex19del in 63.5% of patients, L858R mutations in 34.8% of patients and with other mutations in 1.6% of patients [29]. Previous EGFR-TKI treatments included afatinib, erlotinib, gefitinib and icotinib [29].

4 Tolerability of Aumolertinib

Aumolertinib had a generally manageable tolerability profile in clinical trials in patients with advanced NSCLC with EGFR sensitizing or resistance mutations [27, 29, 30, 32]. In the phase 3 AENEAS trial in treatment-naïve patients with advanced NSCLC and sensitizing mutations, the median duration of treatment was 463 days in the aumolertinib 110 mg once daily arm compared with 254 days in the gefitinib 250 mg once daily arm [32]. Adverse events were reported in similar proportions of aumolertinib (211/214; 98.6%) or gefitinib (213/215; 99.1%) recipients. Grade 3 or higher adverse events were reported in 36.4% and 35.8% of patients in the respective treatment arms, including 20.1% and 26.5% with drug-related grade ≥ 3 adverse events. Serious adverse events (SAEs) occurred in 22.0% of aumolertinib and 21.4% of gefitinib recipients; 4.2% and 11.2%, respectively, experienced drug-related SAEs. Drug-related adverse events that led to treatment interruption occurred in 14.0% of aumolertinib recipients and 24.2% of gefitinib recipients. Drug-related adverse events leading to dose reduction (4.2% and 4.7% in the respective treatment arms) and treatment discontinuation (2.8% and 5.1%) occurred infrequently. One patient in each treatment arm died as a consequence of a drug-related adverse event [32].

The most frequent adverse events (any grade) with aumolertinib in the AENEAS trial were increased blood creatine phosphokinase (CPK), increased aspartate aminotransferase (AST) and increased alanine aminotransferase (ALT), while those in the gefitinib arm were increased ALT, increased AST and diarrhoea. Rash and diarrhoea were less common with aumolertinib (Fig. 3). The most common grade ≥ 3 adverse events (incidence of ≥ 5%) were increased CPK in the aumolertinib arm and increased ALT and AST with gefitinib (Fig. 3) [32]. QT prolongation was reported in 10.7% of aumolertinib and 8.8% of gefitinib recipients; few patients in either treatment group had grade ≥ 3 QT prolongation (0.9% and 1.9%). Interstitial lung disease (ILD) was reported in few patients in either treatment group (0.9% and 0.5%), and none of these events were grade ≥ 3 [32].

In the phase 2 part of the APOLLO trial (Sect 3.2), the most commonly reported treatment-related adverse events with aumolertinib 110 mg once daily (occurring at an incidence of ≥ 10%) were increased CPK (in 20.9% of patients), rash (13.9%), increased AST (12.3%), decreased white blood cell count (12.3%), increased ALT (11.9%) and pruritus (10.7%) [29]. Grade 3 or higher treatment-related adverse events were reported in 16.4% of patients and the most frequent of these were increased CPK (7.0%) and increased ALT (1.2%). Of the 51 patients with elevated CPK, 14 required dose interruptions and three required dose reductions; none required treatment discontinuation and study withdrawal [29]. Fifteen patients (6.1%) in APOLLO experienced QT interval prolongation; none of these patients developed grade ≥ 3 QT prolongation [29]. In all cases, the QT interval normalized without aumolertinib dosage modification. One patient in the trial developed grade 4 heart failure; following study drug discontinuation and symptomatic treatment, the patient recovered. One patient who received aumolertinib 260 mg in the phase 1 part of the APOLLO trial developed ILD; the patient recovered following study drug discontinuation and symptomatic treatment [27]. No patient treated with aumolertinib 110 mg in the phase 2 part of the trial developed ILD [29].

For patients with known cardiovascular risk and conditions which may affect left ventricular ejection fraction, monitoring of cardiac function should be considered [23]. Patients with clinically significant abnormalities in cardiac rhythm or conduction were excluded from the AENEAS [32] and APOLLO [29] trials.

5 Dosage and Administration of Aumolertinib

The recommended dosage for aumolertinib is 110 mg taken orally once daily without regard to food; in patients who have difficulty swallowing solids, contents of the aumolertinib capsule can be dispersed in

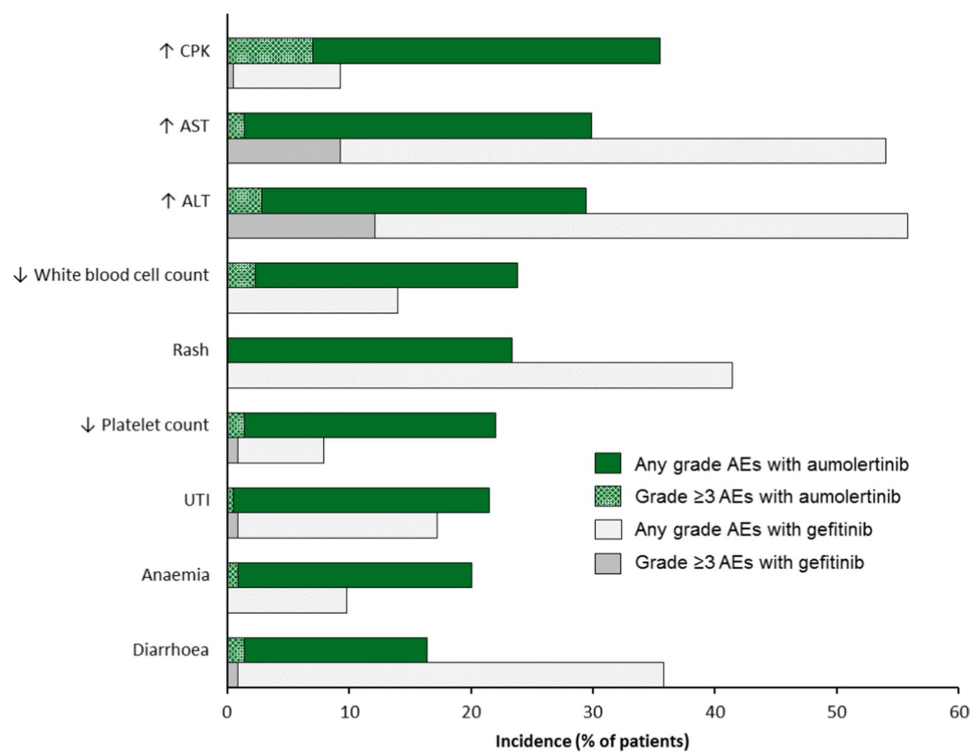

Fig. 3 Tolerability of aumolertinib. Adverse events reported with oral aumolertinib 110 mg (*n* = 214) or gefitinib 250 mg (*n* = 215) once daily as first-line therapy in patients with NSCLC with EGFR-sensitizing mutations in the phase 3 AENEAS trial [32]. ↑ increased, ↓ decreased, *AEs* adverse events, *ALT* alanine aminotransferase, *AST* aspartate aminotransferase, *CPK* creatine phosphokinase, *UTI* urinary tract infection

approximately 50 mL of noncarbonated water and consumed or administered via nasogastric tube [23]. Treatment should be continued until disease progression or unacceptable toxicity occurs. Dosage reduction or interruption can be considered for the management of tolerability issues. Consult local prescribing information for detailed information concerning the administration of aumolertinib in patients who have missed doses, the management of adverse reactions, contraindications, warnings and precautions, and the use of the drug in special populations [23].

6 Place of Aumolertinib in the Management of EGFR Mutation-Positive NSCLC

EGFR mutations are the most frequent genetic mutations in NSCLC, comprising 10–17% of all the mutations in Western patients and approximately 30–50% in Asian patients [5]. Approximately 90% of EGFR mutations in NSCLC are sensitizing ex19del and L858R mutations. Acquired T790M mutation is the most common resistance mutation in patients with sensitizing EGFR mutation-positive advanced NSCLC with disease progression after first-line EGFR-TKI treatment, accounting for 50–60% of cases [33]. Based on its enhanced selectivity for mutated EGFR over wild-type EGFR (resulting in the good efficacy and an improved tolerability profile seen in the FLAURA trial [15, 16]) and intracranial penetration [13, 14, 34], osimertinib is recommended in treatment guidelines as the preferred treatment option as first-line treatment (ahead of earlier-generation EGFR-TKIs) for patients with sensitizing EGFR mutation-positive advanced NSCLC and for patients with EGFR T790M mutation-positive NSCLC that has progressed after first-line EGFR-TKI treatment, including in patients with CNS metastases [35–37].

Like osimertinib, aumolertinib shows potent activity against EGFR harbouring resistance (T790M) and sensitizing (ex19del and L858R) mutations and low activity against wild-type EGFR in vitro and has good intracranial penetration in vivo (Sect. 2). Consistent with in vitro and in vivo activity against both sensitizing and resistance mutations (Sect. 2), aumolertinib has shown good efficacy in clinical trials in patients with advanced EGFR mutation-positive NSCLC (Sect. 3). In the phase 3 AENEAS trial, aumolertinib as first line treatment significantly prolonged median PFS and DoR compared with gefitinib in patients with EGFR mutation-positive (ex19del or L858R) advanced NSCLC (Sect. 3.1). Median PFS and DoR with first-line aumolertinib [19.3 and 18.1 months (Table 1)] are consistent with results seen with first-line osimertinib in the FLAURA trial (median 18.9 and 17.2 months, respectively [16]). OS data for aumolertinib and gefitinib in AENEAS are not mature and final results are awaited with interest. In the phase 1/2 APOLLO trial, aumolertinib showed good efficacy as second-line treatment in patients with advanced NSCLC with evidence of T790M mutations who had progressed after first- or second-generation EGFR-TKI treatment, with a final median OS of 2.5 years in the overall population and ≈ 3 years in the population without CNS metastases at baseline (Sect 3.2).

Aumolertinib was also effective as first- or second-line treatment in subgroups of patients with CNS metastases at baseline (Sect. 3). PFS was significantly prolonged in the aumolertinib arm compared with the gefitinib arm in this group of patients in AENEAS (Sect. 3.1). In APOLLO, ORR, DCR, PFS and DoR in those with CNS metastases were comparable with those achieved in the overall population, and the median OS at final assessment was ≈ 1.5 years (Sect 3.2). The efficacy of aumolertinib in patients with EGFR mutation-positive advanced NSCLC who have CNS metastases is being investigated further in the ARTISTRY trial, a dose-exploration study of first- or second-line treatment in patients with newly diagnosed or recurrent brain or leptomeningeal metastases [38].

Aumolertinib had a generally manageable tolerability profile in clinical trials in patients with advanced NSCLC with EGFR sensitizing or resistance mutations (Sect. 4), consistent with that of gefitinib. However, adverse events associated with inhibition of wild-type EGFR, such as rash and diarrhoea, were less frequent with aumolertinib than gefitinib (Sect. 4). In contrast, only rash was less frequent with osimertinib when compared with standard EGFR-TKI treatment (gefitinib or erlotinib) in the FLAURA trial [16]. Most adverse events with aumolertinib were of mild to moderate severity.

As yet, there are no direct or indirect comparisons between aumolertinib and second- or third-generation EGFR-TKIs; such studies would be of interest. Nevertheless, based on good efficacy outcomes and a generally manageable tolerability profile, aumolertinib is a promising new first- or second-line treatment option in patients with advanced EGFR mutation-positive NSCLC, including in patients with CNS metastases.

Data Selection Aumolertinib: 172 records identified	
Duplicates removed	0
Excluded during initial screening (e.g. press releases; news reports; not relevant drug/indication; preclinical study; reviews; case reports; not randomized trial)	0
Excluded during writing (e.g. reviews; duplicate data; small patient number; nonrandomized/phase I/II trials)	136
Cited efficacy/tolerability articles	5
Cited articles not efficacy/tolerability	31
Search Strategy: EMBASE, MEDLINE and PubMed from 1946 to present. Clinical trial registries/databases and websites were also searched for relevant data. Key words were almonertinib, aumolertinib, Ameile, HS10296, NSCLC. Records were limited to those in English language. Searches last updated 14 February 2022.	

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Declarations

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Ethics approval, Consent to participate, Consent to publish, Availability of data and material, Code availability Not applicable.

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